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Rethinking On-Policy Distillation: One Training Example

A Single Query Recovers Most of Full-Dataset OPD Gain — Revealing That LLM Distillation Is Algorithm-Starved, Not Data-Starved

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On-policy distillation (OPD) of large language models is widely believed to require large, diverse training datasets. This paper refutes that assumption: training OPD on a single query recovers most of the accuracy gain achieved by full-dataset training on mathematical reasoning, code, and general benchmarks across multiple model families. The finding reveals that OPD is *data-overfed but algorithm-starved* — the bottleneck lies in the optimization procedure, not the volume of training queries.

AT A GLANCE

Problem: OPD is assumed to require large diverse datasets; no theory or experiment had tested the data-minimal limit.

Why it matters: If one query suffices, OPD can be applied at very low data cost, democratizing teacher-guided alignment.

Method: Fix training steps; vary query count from 1 to the full dataset; measure accuracy and state coverage.

Result: One query achieves 71.5% state coverage and most of full-data OPD’s gain; 16 queries reach 98.9% and match full data.

Evidence: Experiments across MATH-500, GSM8K, LiveCodeBench, MMLU-Pro with Qwen2.5-7B, Llama-3.1-8B, DeepSeek-R1-Distill-7B.

The Big Picture

On-policy distillation combines student-generated rollouts with dense token-level supervision from a teacher LLM. The student generates responses, the teacher evaluates them, and the student updates its parameters using those signals — with rollouts regenerated at each iteration from

the updated student.

The conventional wisdom is that OPD needs a large and diverse set of training queries to cover the space of possible inputs and to produce varied rollouts for learning from. This view motivates the standard practice of assembling large question banks for distillation.

This paper asks a simple question: what happens at the data-minimal limit, i.e. when OPD trains on exactly one query? The answer, that one-shot OPD keeps improving for hundreds of steps and recovers most of full-data OPD’s gain, reframes the problem of LLM alignment distillation entirely.

Why Existing Approaches Fall Short

Current OPD practice assembles datasets of thousands of training queries (e.g. 512–4096 mathematical problems), under the assumption that broad coverage is necessary for generalization. This creates two problems:

First, large datasets impose *data collection costs*: curating high-quality, diverse questions for every new target domain requires significant effort.

Second, large datasets may *mask algorithmic limitations*: if the training data is the bottleneck, then improving the algorithm yields little gain unless data is first increased. This paper reveals that the data is *not* the bottleneck, so algorithmic improvements (better teachers, better update rules, better rollout strategies) are the correct direction for future work.

Core Mechanism

The key insight is that even with a fixed single query q^* , the rollout distribution $\pi_\theta(\cdot \mid q^*)$ changes at every gradient step as θ is updated. This *dynamic rollout diversity* provides gradient signal across different student behaviours even when the input prompt is fixed.

The authors measure this via **state coverage**: the fraction of intermediate rollout states visited by a given query set that are also visited by the full dataset:

$$\text{Coverage}(Q) = \frac{|S(Q) \cap S(Q_{\text{full}})|}{|S(Q_{\text{full}})|}$$

One query achieves Coverage = 71.5% — most of the state space is reachable from a single starting point as the model evolves.

Technical Formulation

The OPD training objective is a token-level imitation loss:

$$\mathcal{L}_{\text{OPD}}(\theta) = -\mathbb{E}_{q \sim \mathcal{D}} \mathbb{E}_{y \sim \pi_\theta(\cdot \mid q)} \sum_t \log \pi_{\text{teacher}}(y_t \mid q, y_{<t})$$

where π_θ is the student, π_{teacher} is the frozen teacher, and y is a student-generated rollout.

At the single-query limit with query q^* , the gradient at step k is:

$$\nabla_\theta \mathcal{L} = -\mathbb{E}_{y \sim \pi_\theta(\cdot | q^*)} \sum_t \nabla_\theta \log \pi_\theta(y_t | q^*, y_{<t}) \cdot \log \pi_{\text{teacher}}(y_t | q^*, y_{<t})$$

Despite the fixed q^* , the expectation is over the current student rollout distribution, which shifts every step — providing effective gradient diversity.

4.1 Coverage Saturation

Empirically, coverage $C(N)$ as a function of query count N follows an approximate power law:

$$C(N) \approx 1 - \alpha N^{-\beta}$$

with fitted constants α, β . Empirically, $C(1) \approx 0.715$ and $C(16) \approx 0.989$.

Learning or Inference Procedure

The authors fix the total training budget (number of gradient steps) and vary only the number of unique training queries. For each query count $N \in \{1, 4, 16, 64, 512\}$:

1. Uniformly sample (or select by diversity) N queries from the training pool.
2. Run OPD for the fixed number of steps on those N queries.
3. Evaluate on held-out benchmarks (MATH-500, GSM8K, LiveCodeBench, MMLU-Pro).
4. Measure state coverage by intersecting rollout state sets.

Query selection at $N > 1$ uses a semantic diversity criterion (maximum marginal relevance on embedding space) to maximise coverage gain.

What the Guarantee Says

The paper does not prove formal guarantees but establishes an empirical regularity: OPD accuracy as a function of query count follows the same coverage power law. Because accuracy and coverage co-move, maximising *semantic diversity* of a small query set — rather than maximising dataset size — is the correct strategy for efficient OPD.

The corollary is that OPD improvements will be unlocked by better algorithms (improved rollout strategies, better acceptance rules, curriculum over query difficulty) rather than by larger datasets.

Experimental Findings

Accuracy vs. query count (Qwen2.5-7B, MATH-500):

Queries	MATH-500	Coverage
1	68.4%	71.5%
4	70.1%	89.2%
16	71.8%	98.9%
512	72.3%	100%

The accuracy gain from 1 query to 512 queries is only 3.9 percentage points, while the gain from the pre-OPD baseline to 1-query OPD is substantially larger — confirming that most of the alignment benefit comes from the OPD mechanism itself, not the data diversity.

Results are consistent across Llama-3.1-8B and DeepSeek-R1-Distill-7B student models, and across GSM8K, LiveCodeBench, and MMLU-Pro benchmarks.

Ablations and Interpretation

Random vs. semantic query selection: At $N = 16$, randomly sampled queries underperform semantically diverse queries by 1.7 percentage points (MATH-500). Coverage is the active ingredient.

Step budget vs. data size: One-shot OPD requires $\sim 2\times$ more gradient steps to match full-data OPD accuracy at $N = 16$, but converges to the same value — showing the algorithm can compensate for data sparsity given sufficient compute.

Teacher quality: Weaker teachers (same-size models) yield diminished but still positive one-shot gains. The mechanism does not require a very large teacher.

Alignment pace: Plotting validation accuracy vs. steps for $N = 1$ and $N = 512$, both curves plateau at the same rate — confirming the algorithm bottleneck hypothesis.

Reference

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